Project

Air flow Analysis over an Ahmed Body

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Objective

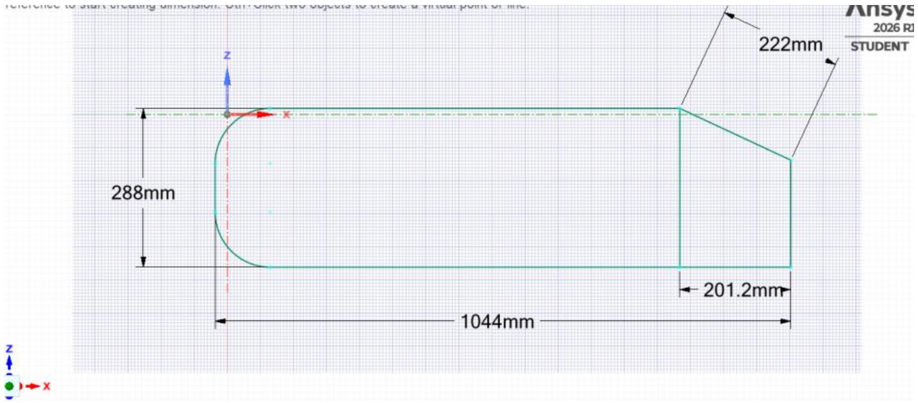
- Objective of this study is to prepare a simplified car geometry, well known as 'Ahmed Body', then mesh and perform a through analysis of air flow including the Drag Coefficient, CD of the Ahmed body in the virtual wind tunnel.

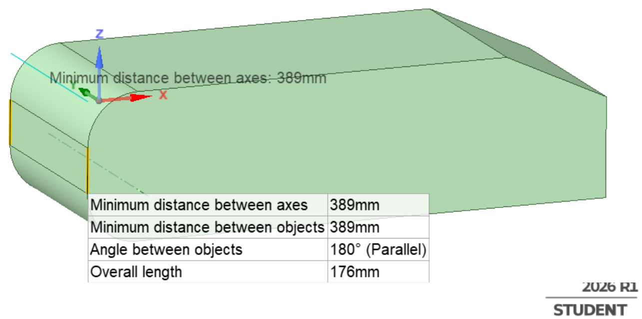
CAD preparation

- A 25 degree Ahmed body is designed. Despite flow separation happening on the later part of the 25 degree slant, compared to the 35 degree slant, two counter rotating longitudinal vortices are dominant throughout the entire flow structure(1). As a result the drag is larger and the flow characteristic is more interesting. Thus 25 degree Ahmed model is chosen

- The dimensions are similar to the standard Ahmed body with a 50 mm ground clearance as Jie Tian et al. Discussed.

- The wind tunnel designed here is significantly smaller than in the article of Jie Tian et al due to computational limitations. The inlet of the outlet is 2 times of its own length aways from the front of the body and the outlet is 4 times away in this simulation. Similarly wind tunnel dimension is reduced on top and sides.


222mm
288mm
201.2mm
1044mm
2026 R1
STUDENT


Minimum distance between axes: 389mm
Minimum distance between axes 389mm
Minimum distance between objects 389mm
Angle between objects 180° (Parallel)
Overall length 176mm
2026 R1
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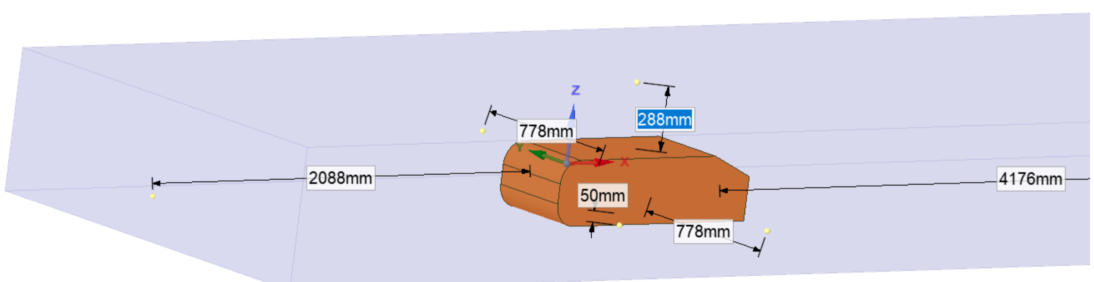
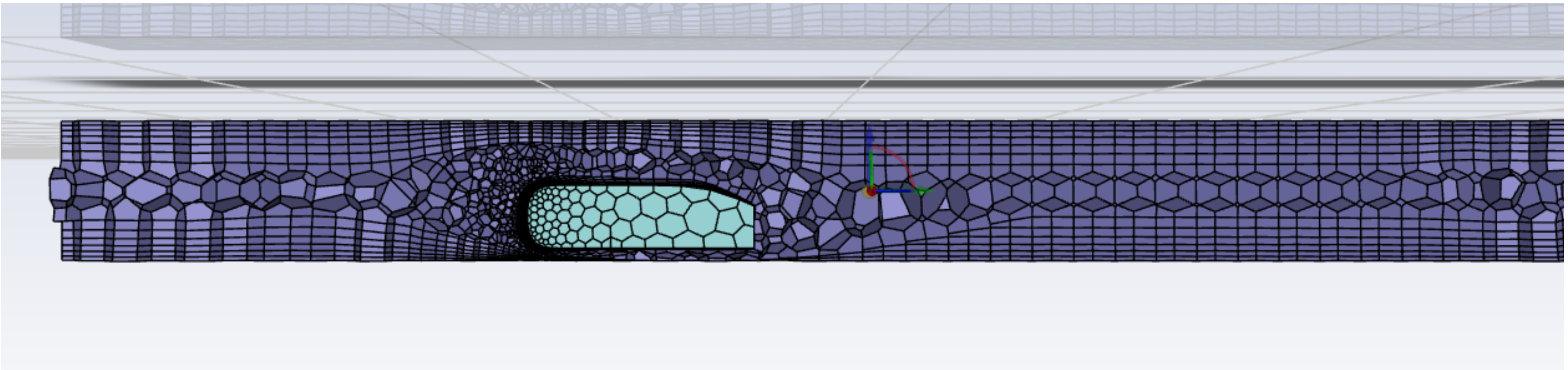

778mm
288mm
2088mm
50mm
778mm
4176mm

Blocking

- CAD was meshed in fluent mesher. Watertight geometry was followed.
- Ahmed body was considered **dead region.**
- 6 boundary layers were created
- Polyhedra was used for this unstructured meshing
- After the meshing was complete, **cell count: 17174**

  Nodes: 50156



PC specifications

- Intel core i5-1135G7 2.40 GHz CPU(4 cores)
- RAM: 8 GB ddr5

CFD Setting

- Parallel processing was implementing using 3 processes
- Quality check was performed. It had orthogonal quality of 0.1355 and aspect ratio of 65.04
- Energy equation was turned off, SST k-omega(2 eqn) model was used for turbulence

  Inlet Velocity : 60 m/s, turbulence intensity: 0.5% and TVR: 6

  Outlet pressure : 0 Pa

Walls were stationary, no slip and adiabatic

SIMPLE scheme was used for solving

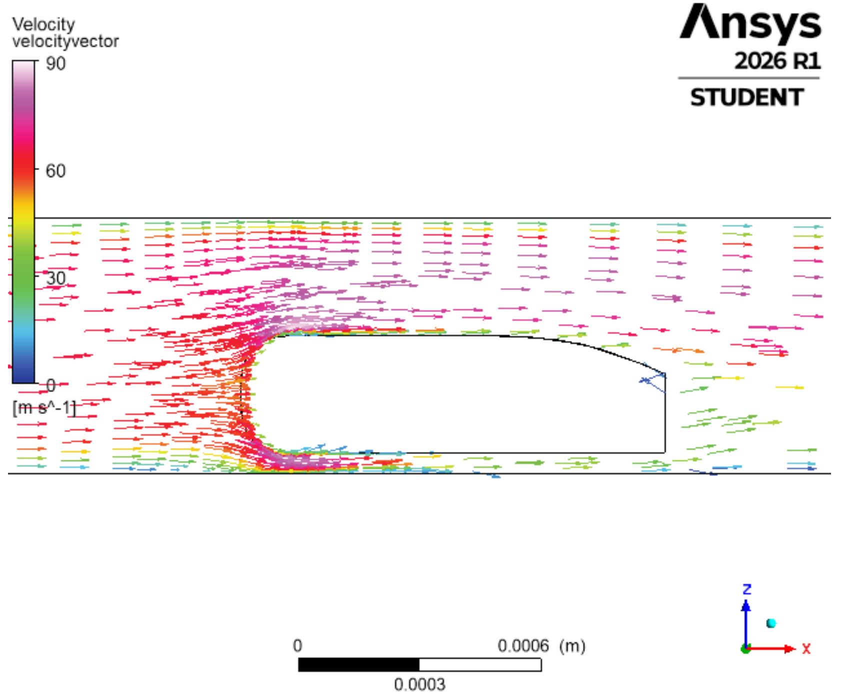
1000 iterations were performed

Results

CFD post was used for post processing

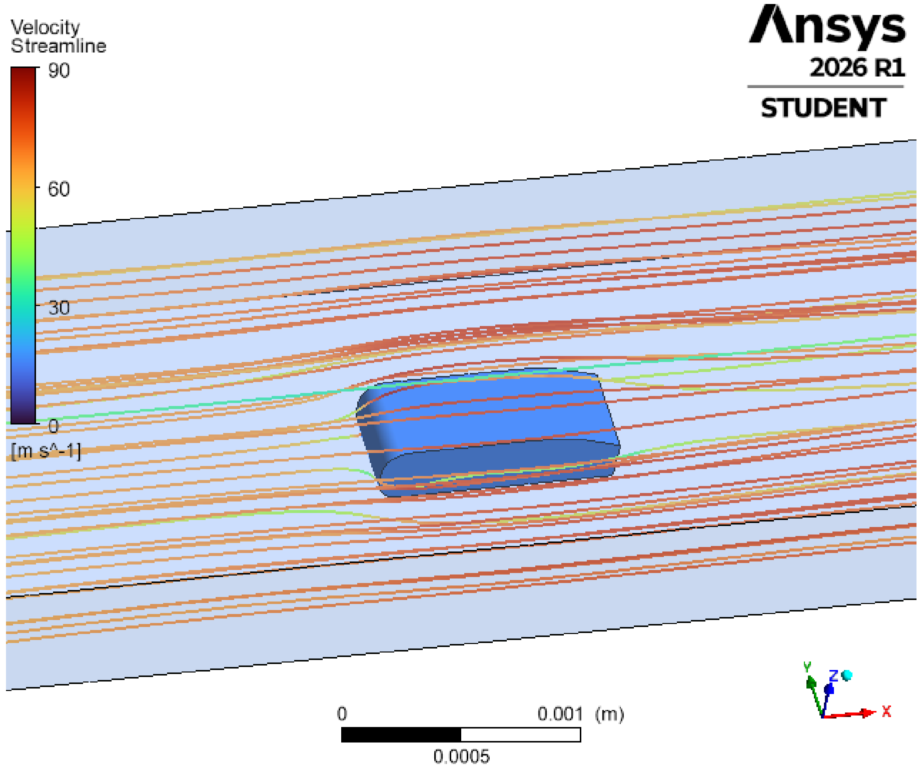
Co-efficient of drag (CD) was calculated using the expression $CD = \frac{F_d}{\frac{1}{2}\rho A v^2}$, where F_d is the drag force, *A* is the area of the Ahmed body, and *v* is the inlet velocity of the fluid flow. CD value was unfortunately 0.0923 in this simulation which is far off from the expected value of CD for a 25 degree Ahmed body. **Probable causes are in Discussions section.**

Velocity vectors



FlowThermoLab

Streamline


Velocity
Streamline
90
60
30
0
[m s^-1]
Ansys
2026 R1
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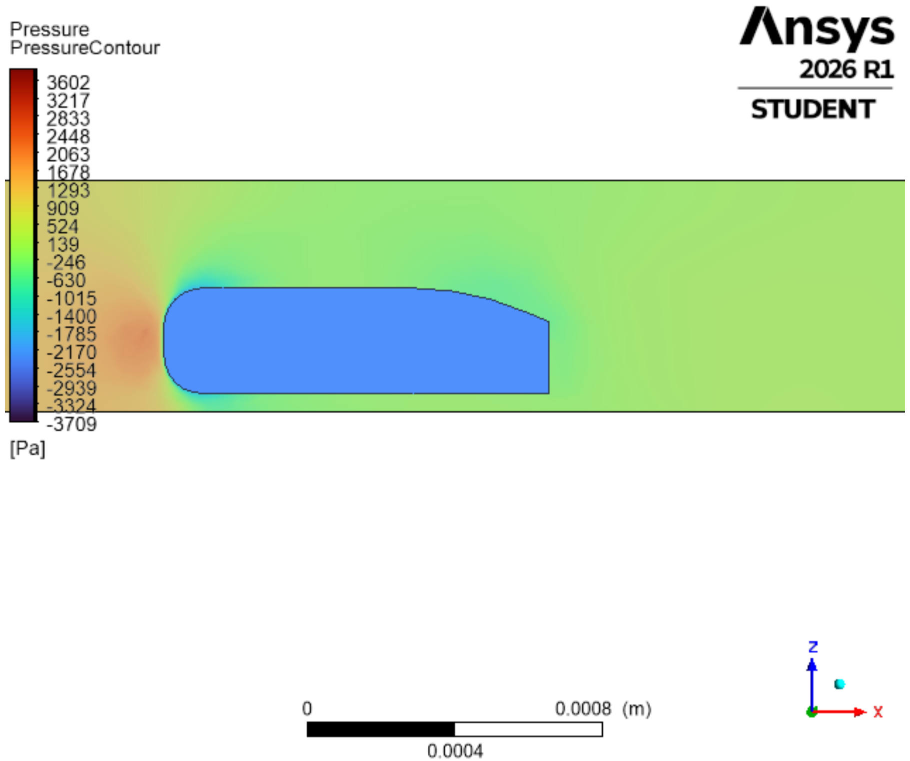
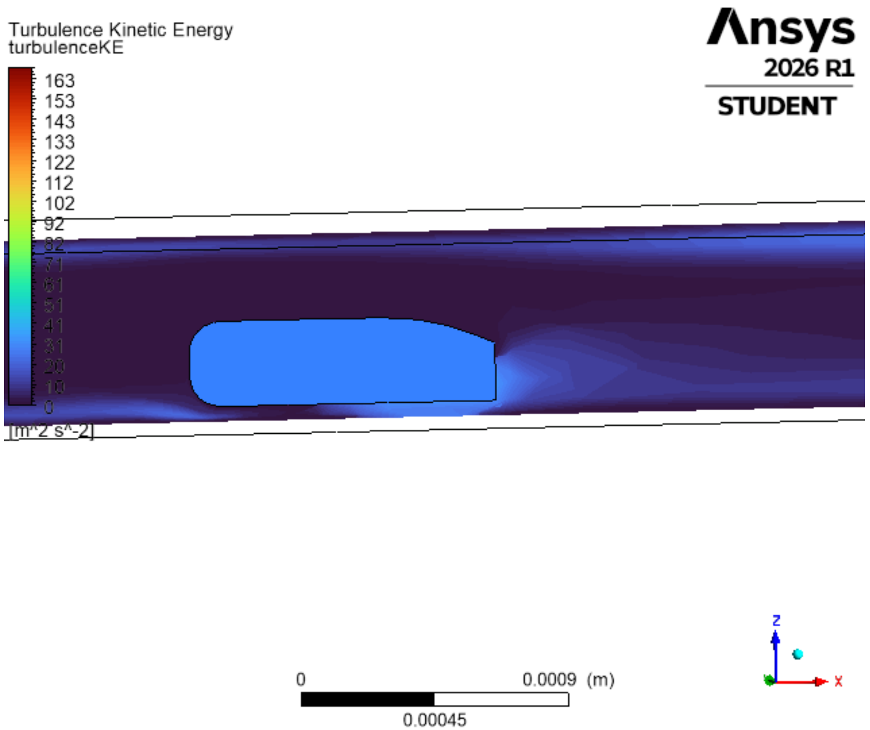
0
0.001 (m)
0.0005
Y
Z
X

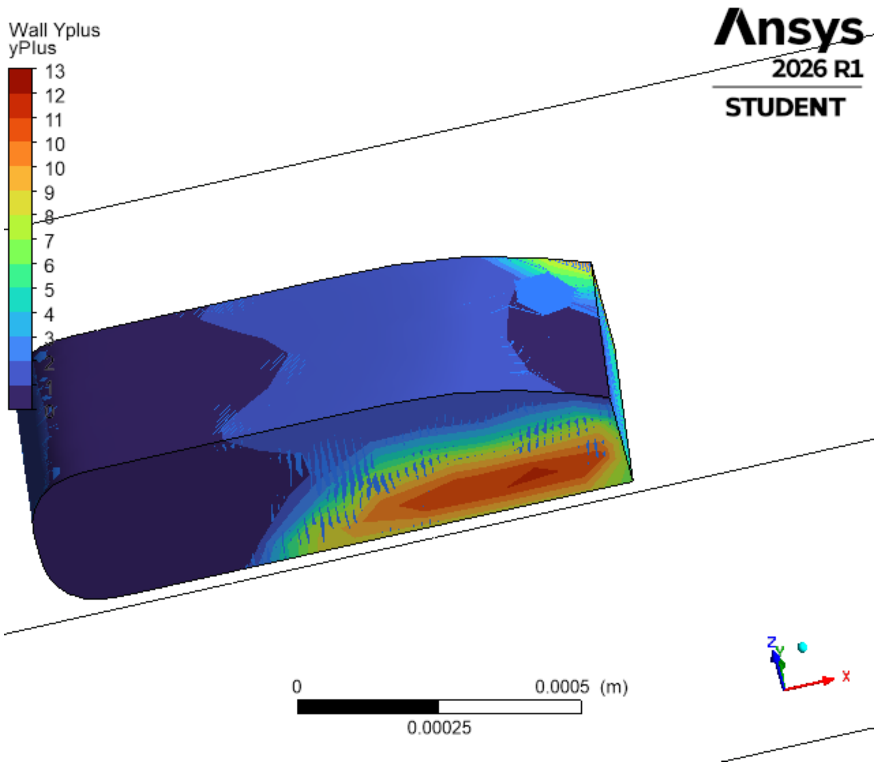
• Pressure Contour



Turbulent KE contour:



y+ contour:


Wall Yplus
yPlus
13
12
11
10
10
9
8
7
6
5
4
3
Ansys
2026 R1
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0
0.0005 (m)
0.00025

Discussion

It is clear from the pressure contour that there in pressure difference at the front and the back of the body. Front is high pressure, back is low pressure region creating pressure gradient and thus contributing to pressure drag.

It is also visible from the turbulent KE contour that there is significant turbulent KE at the back of the body.

The deviation from the expected value(0.25 ish) may be the result of insufficient wind tunnel size.

References

(1). Jie Tian, Yingchao Zhang, Hui Zhu and Hongwei Xiao, Aerodynamic drag reduction and flow control of Ahmed body with flaps